

R Sai Pranav

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Profile Summary

Computer Science and Engineering student and Software Development Engineer (SDE) Intern with hands-on experience in backend development, full-stack web development, and scalable system design. Ships production code: REST APIs in live manufacturing at Cepheid, the backend of a live AI SaaS product, and 50+ merged open-source patches to C codebases including PostgreSQL tooling. Strong in data structures and algorithms, debugging, and testing.

Education

REVA University, Bachelor of Technology in Computer Science Engineering

2023 – Present

Bengaluru, Karnataka | GPA: 8.54/10.0

Experience

Software Development Engineer (SDE) Intern, MES

Sep 2026 – Present

Cepheid India Pvt Ltd (Danaher Corporation) – Bengaluru

- Develop and maintain REST Web APIs for integration on Opcenter (Siemens MES), customized for business needs, using SQL Server and AWS infrastructure.
- Ship in monthly Agile sprints from user stories to deployment, maintaining code used in live manufacturing.

Backend Developer (Freelance)

May 2026 – Sep 2026

Contralyne, AI contract-review SaaS (contralyne.com) – Remote

- Built the Node.js, Express, and TypeScript backend for a live product built for US and UK legal teams: clause-level risk analysis and redlined DOCX export with tracked changes.
- Owned the data and security layer (Supabase PostgreSQL, AWS S3 with AES-256, JWT authentication, per-route rate limiting); top contributor with 225 commits.

Software Engineering Intern, Full Stack

Dec 2025 – Aug 2026

MyStartupWave – Bengaluru (Hybrid)

- Built and shipped 10+ full-stack web applications using React, FastAPI, Node.js, and MongoDB from scratch to production.
- Cut API response times by up to 35% through performance fixes and caching, backed by unit, integration, and automated API testing.

Project Intern, Cybersecurity Engineering

Nov 2025 – Aug 2026

iSPIRT Foundation (Sakram) – Remote

- Worked on protocol frameworks for India's Digital Public Infrastructure (DPI) in C++ and Python (hardware-bound cryptographic validation, data integrity); fixed 7 failing integration tests and removed exposed database credentials.

Research Intern, Explainable AI

Jul 2025 – Jan 2026

CSIR-4Pi (Fourth Paradigm Institute) – Remote

- Built a PyTorch breast-cancer segmentation pipeline (MedSAM, ResNet, U-Net) with XAI heatmaps; co-authoring a research paper.

Open Source Contributions

- 50+ patches merged into PostgreSQL-ecosystem tools written in C (pgmoneta 22, pgagroal 15, pgvictoria 7) and into GRASS GIS, JabRef, GNU Radio, and kubernetes/website, with 20+ more PRs in review.
- Found and fixed memory-safety defects in C — including a stack-buffer overflow from an unbounded strcat, unchecked realloc crashes, and use of freed pointers — and stabilized flaky CI tests in GNU Radio and JabRef.
- Two security fixes (in review) in interviewstreet/hiring-agent (7k+ stars, HackerRank OSS): blocked prompt injection where PDF resumes hide LLM-manipulating text in background-colored ink, and via forged section delimiters.

Projects & Achievements

claude-orc — Open-Source Python CLI, Published on PyPI (pip install claude-orc)

- Sole-authored a CLI for Claude Code users across multiple accounts and machines: git-backed memory sync, a skill registry, and token-usage reports parsed from local transcripts; 160+ tests run in GitHub Actions on Python 3.10–3.13, and CI publishes releases to PyPI.

Evidence Verifier — Razorpay AI Buildathon

- Verifies invoices and dispute evidence with cryptographic and arithmetic checks (signed QR codes, GST e-invoice payloads, GSTIN checksums) instead of pixel forensics; zero false positives across 9 independent test corpora; logged and fixed 26 defects, most found by testing on real documents.

Mortgage Document Intelligence Pipeline — InfrX 2026, 2nd Place (₹75,000)

- Classified 100–2,000-page mortgage packages into 27 document types at 91% accuracy; a cost-aware cascade (pdfplumber, then heuristics, then a vision model) keeps most pages away from the LLM.

HackerRank Orchestrate (Aug & Sep 2026) — Two Solo Builds

- WhatsApp notification router whose deterministic tier settles 23.6% of messages without a model call (run cost cut from \$2.80 to \$0.76), and a 90-day cash-flow affordability engine with 38 tests.

Decentralized Spatial-Temporal Data Veracity System — Personal Patent, Filed 2026

Technical Skills

Languages: Java, Python, C, C++, JavaScript/TypeScript, SQL, HTML/CSS

Technologies: Node.js, Express, FastAPI, React, PostgreSQL, MongoDB, SQL Server, Docker, Git, Linux, Postman

Cloud: AWS, Google Cloud

Concepts: Data Structures & Algorithms, Debugging, OOP, REST APIs, Microservices, System Design, Unit & Integration Testing, CI/CD, Code Review, Agile/Scrum, Observability & Logging

Leadership & Involvement

- National Winner, Cardano Hackathon Asia (200+ teams); Finalist, Google DeepMind Hackathon; Top 5, AWI SpaceTech Hackathon.
- Technical Lead, Algorand Club, REVA University (2026–Present): organized 5+ hackathons and competed in 20+.